# Adi Raj

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## Education

**Harvard University** Cambridge, MA
*A.B./A.M. in Mathematics (Honors) and Physics - GPA 3.91* *September 2023 - May 2027*

- **Coursework:** MATH55a/b, differential geometry, commutative algebra, algebraic geometry, algebraic topology, markov chains, evolutionary dynamics, complex dynamics, physical mathematics, fundamentals of algebraic data representation, quantum mechanics, statistical mechanics, general relativity, nonlinear algebra

## Experience

**Harvard SEAS Applied Algebra Group** Cambridge, MA
*Research with Prof. Anna Seigal* *September 2025 – Present*

- *Causal representation learning:* Developing identifiability proofs for unpaired multi-modal data using perfect intervention data with overcomplete ICA; showed one perfect intervention per latent node across modes is sufficient and worst-case necessary to recover latent directed acyclic graphs and mixing matrices
- *Tensor methods:* Developing theory of real points in complementary sections of determinantal varieties, using probabilistic and topological methods to study matrix-space ranks and data tensor decomposition identifiability

**Harvard University Math Department** Cambridge, MA
*Thesis Research with Prof. Joe Harris* *January 2026 – Present*

- Analyzing Schubert problems on orthogonal Grassmannians of Lie types B and D using Harris' monodromy method for determining Galois groups, Vakil's geometric Littlewood–Richardson rule, and Schubert induction
- Studying enumerative structure of isotropic subspaces on Lagrangian Grassmannians using intersection theory

**Trinity College, University of Cambridge** Cambridge, UK
*Visiting Research with Prof. Rebecca Fitzgerald* *May 2025 - August 2025*

- Analyzed bulk RNA-seq and whole-genome sequencing data to quantify transcriptomic and genomic determinants of tumor microenvironment influencing esophageal cancer progression
- Used survival modeling and multivariate association testing to identify statistically robust prognostic biomarkers

**Center for Astrophysics | Harvard & Smithsonian** Cambridge, MA
*Research with Prof. Rosanne di Stefano* *June 2024 - March 2025*

- Designed scalable statistical inference pipelines for modeling temporal properties of 400,000+ CSC X-ray sources
- Developed statistical inference methods for sparse time series to recover latent temporal structure in irregular data

**Harvard University** Cambridge, MA
*Course Assistant for algebraic geometry, topology, analysis, group theory, theoretical physics* *August 2024 - Present*

- Holding weekly office hours, exam review, and sections teaching rigorous proof-based classes
- Grading homework proofs and providing detailed individual assistance and feedback

## Activities

**Harvard Fencing Club** (Captain): Leading sabre training, managed funding, season planning, and tournament operations
**Harvard Chess Club**: Investigating learned representations in Stockfish NNUE through competitive chess

## Awards & Fellowships

**Program for Research in Science and Engineering (PRISE) Fellow**: Supported representation learning research
**David Rockefeller International Experience Fellow**: Supported research at Trinity College, University of Cambridge
**Harvard College Research Program (HCRP) Fellow**: Supported multidisciplinary independent scholarly work
**John Harvard Scholar**: Awarded to top 5% of Harvard College students

## Volunteering

**Cambridge Math Circle**: Supported mathematical enrichment and problem-solving activities for local K-8 students
**Volunteer Tutoring**: Taught Geometry, Algebra II, Precalculus, and Physics to local high school students
**Harvard Square Homeless Shelter**: Helped make meals, clean utensils, and facilitate a safe, clean, and caring space